APRIL 2, 2026

Matt | Varghese Mathew

Senior Software Engineer

Senior Software Engineer with 15+ years of hands-on experience designing large-scale distributed platforms and cross-platform regulatory infrastructure. Meticulously translates evolving specifications into production systems adopted across hundreds of enterprise customers. Recognized for unifying complex multi-system integrations into scalable architectures through deep technical research and rigorous system design.

Ask Me || DM me on LinkedIn

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Madison, WI

linkedin.com/in/matt-varghese

github.com/mattvarghese

WORK EXPERIENCE

Senior Software Engineer (L4), Utilization Management

Epic - Verona, WI — *MAR 2021 - PRESENT*

Architect for a large-scale, standards-based, regulatory integration platform, defining automated real-time decision workflows for enterprise-scale organizations. Engineered systems adopted by hundreds of major enterprise clients, reducing high-latency business process turnarounds from several days to near real-time.

- **Led architecture and delivery of a regulatory interoperability platform** for real-time data exchange, collaborating with standards bodies and external vendors to transform emerging specifications into scalable, production-ready infrastructure.
- **Identified gaps and established a unified "source of truth"** replacing fragmented logic across 12+ product modules, reducing lead time for new feature development by ~80%, and improving long-term maintainability.
- **Recognized the need and architected standardized** data models and execution frameworks replacing inconsistent one-off implementations, significantly improving maintainability and cross-team scalability.
- **Modernized third-party decision-engine integrations** using OAuth2 SSO patterns, enabling real-time workflow execution and reducing integration defects by ~60%.
- **Mitigated tooling deficiencies by building simulation and testing frameworks** to emulate third-party systems, enabling end-to-end workflow validation, proactively eliminating defects, and accelerating partner onboarding.

Software Engineer (L2, L3 2015), Clinical Informatics

Epic - Verona, WI — *AUG 2010 - MAR 2021*

Senior developer responsible for distributed decision-support infrastructure, external web application integrations, and mobile push notifications services supporting large-scale enterprise applications. Directed adoption of modern interoperability standards and pioneered Epic's first high-availability cloud service on Azure.

- **Created Epic's high-availability mobile push notifications platform,** routing millions of weekly notifications via APNS and GCM. Pioneered deployment as one of Epic's first Azure-native production services, reducing operational costs by over 90%.

SKILLS

Engineering Stack

- TypeScript • React • Web API
- Node • .NET/C# • C/C++
- Python • Git & SVN • ANSI-M
- SQL & NoSQL • Linux & Windows

Architecture & Infrastructure

- SDLC • Distributed Systems
- Microservices • Azure • CI/CD
- Docker • Virtualization • Hadoop
- Maintainability & Performance

Specialized Expertise

- LLM API Integration
- Agentic Workflows
- Regulatory Infrastructure
- Network Architecture & Routing

COLLABORATION

Standards-body Hackathons

Developed prototypes of industry specifications in collaboration with participants from other vendors to improve interoperability.

Authoring Industry Standards

Contributed to two emerging industry specifications for interoperability towards improving workflow integration & security.

- **Identified fragmentation and led enterprise-wide centralization** of 3rd-party web application integrations to establish standards for security and maintainability. Developed a suite of validation apps for CSP, CORS, and Web Messaging APIs.
- **Designed & Developed Epic's decision support web services framework,** enabling reusable standards-based workflow integrations across multiple enterprise applications using RESTful APIs and event-driven execution patterns.
- **Identified usability gaps and redesigned** decision support configuration to align with mental models, improving usability and enabling efficacy feedback loops for users.
- **Subject matter expert in Web App Security and RESTful integrations,** providing architectural guidance to multiple teams and onboarding external vendors.

KEY PROJECTS

LLM-Integrated Workflow Assistant *2025*

Designed and implemented a single-sign-on AI assistant web application. App interfaces LLM APIs with structured enterprise data sources to support real-time chat-based interaction for in-workflow assistance. Demonstrates applied architecture for AI-based workflow orchestration over live API-backed data sources.

RFC-compliant JSON Serializer & Deserializer for ANSI-M *2020*

Created RFC 8259 compliant generic JSON Serializer and Deserializer for the ANSI-M language. Library enables 30+ database AI integrations and analytics across Epic. Initiative spotlights architectural vision in creating powerful generic reusable tooling.

MIPS Microprocessor Pipeline Simulator *2005, 2025*

github.com/mattvarghese/coconut

Demonstrates expertise in computer architecture, parallelism, and performance. 5-Stage pipeline simulation of a MIPS microprocessor, with cache simulation, assembler, and terminal app developed with POSIX APIs, C++, Flex, and Bison in 2005. In 2025, developed a Small-C compiler above the assembler with AI assistance.

PRIOR EXPERIENCE

Research Assistant, Condor Distributed Computing

University of Wisconsin - Madison, WI *SEP 2008 - MAY 2010*

Virtualized Condor jobs with VMware achieving deterministic runtime environments.

Software Engineer, Enterprise Routing

Cisco Systems - Bangalore, India *JUN 2006 - JUL 2008*

Platform engineering and configuration tooling for Cisco's enterprise class routers.

EDUCATION

Master of Science, Computer Science

University of Wisconsin - Madison *2008-2010*

GPA: 3.94 / 4.0 | Focus: Systems, Networking, Distributed Computing.

Bachelor of Technology, Computer Science and Engineering

National Institute of Technology - Calicut, India *2002-2006*

CGPA: 9.60 / 10.0 | Department Topper and Gold Medalist.

Steering Industry Standards

Represented Epic on public calls steering development of three standards to reduce administrative overhead for high-latency B2B transaction approvals.

Epic Vendor Conferences

Presented for several years on Epic's web service integration capabilities to educate and onboard third-party vendors.

Epic Mentoring

Mentored engineers and interns across multiple teams, driving adoption of architectural best practices and development of high-performing teams.

Epic Recruitment

Attracted top talent to Epic by representing Epic at university career fairs and conducting candidate interviews.

EXPLORATORY

Web Based SSH Client *2025*

Non-traditional solution for real world cost reduction avoiding remote-app licenses for moderated shell access to servers.

Demonstrated proof of concept fully browser-based SSH client for unix servers.

github.com/mattvarghese/WebSSHClient

HTTP Auto-responder *2024*

Development tooling for Web APIs, keeping traffic within the enterprise or local network.

Configurable endpoint allows simultaneous multi-user testing of RESTful APIs.

github.com/mattvarghese/HttpResponder

OTHER

- USHPA Novice (H2) rated hang-glider pilot
- Solo endorsed VFR student pilot
- ASA 101,103/104,105 rated sailor
- Learned Elementary Russian as an adult
- Avid reader, ice-skating, volleyball, XC-skiing, bicycling